

Focused Primary-Evidence Audit: Human CFAP410 (O43822) Subcellular Localization for GO Curation

Date: September 2026 **Scope:** Claim-by-claim provenance audit of GO cellular-component annotations for human CFAP410 (UniProt O43822; gene *C21orf2* / *LRRC76*; Ensembl ENSG00000160226; mouse ortholog Q8C6G1), with emphasis on the disputed plasma-membrane IDA cited to Bai et al. 2011.

Summary

The central finding of this audit is that the **CFAP410 GO:0005886 "plasma membrane" IDA annotation, cited to Bai et al. 2011 (PMID: 21834987, doi:10.1186/1741-7007-9-54), is a curation error and should be removed.** A full-text audit of the paper — including all 14 supplementary files — shows that C21orf2/CFAP410 was never expressed as a tagged protein, and its subcellular localization was never directly assayed. The protein appears in Bai2011 only as an RNAi knockdown hit that produced a *cortical actin* morphology phenotype (Table 2). The paper's sole localization experiment (GFP tagging, Figure 8) covered five entirely different proteins — ARC, FAM40A, FAM40B, FMNL3, and ZRANB1 — which showed "cytoplasmic localization with some enrichment on the plasma membrane." The plasma-membrane IDA for CFAP410 was thus mis-derived, most likely by conflating (a) the phenotype-class definition "cortical actin = increased F-actin on regions of the plasma membrane" with (b) the unrelated GFP-fusion result for the five other proteins. The 2024 Author Correction (PMID: 39390519) fixes only a duplicated figure panel and does not touch C21orf2 or any localization claim.

Corroborating this verdict, **UniProt's manually curated Subcellular Location comment for O43822 neither lists plasma membrane nor cites Bai2011.** The plasma-membrane IDA is therefore an orphan GOA annotation, inconsistent with the expert-curated record. The same Bai2011 citation is also attached to a GO:0005737 "cytoplasm" IDA; that cytoplasm term nonetheless survives independently because it has a second reference (Fang et al. 2015, PMID: 26290490), whose exact experimental basis could not be verified here because the full text was inaccessible.

The **best-supported localization for CFAP410 is the ciliary base** — the basal body, daughter basal body, connecting cilium, and transition zone. This is backed by direct human/pig/mouse retinal immunohistochemistry (Khan 2015), an independent ciliopathy screen (Whewey 2015), direct human iPSC-derived motor-neuron imaging (De Decker 2025), and the Human Protein Atlas. In contrast, the **mitochondrial** annotation (Krohn 1997) rests on 1997-era polyclonal antisera without modern isoform or knockout controls, and the **photoreceptor outer segment** GO term (Suga 2016) is over-specified relative to the paper's own wording ("photoreceptor cilia"). Neither the NEK1–C21ORF2 physical association nor any other protein–protein interaction supports a molecular-function, scaffold, or additional-localization claim on its own.

Key Findings

F001 — The plasma-membrane IDA (GO:0005886) from Bai2011 is unsupported

QuickGO records O43822 GO:0005886 "plasma membrane" as an **IDA** annotation with reference

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(assignedBy UniProt); the same paper is also the IDA source for GO:0005737 "cytoplasm." A full-text audit of Bai2011 (PMC3201212) shows that **C21orf2 appears only twice in the entire article**: (1) in a gene-name table, and (2) in Table 2 as an RNAi knockdown morphology phenotype recorded as "C21orf2 — cortical actin — 55%."

The paper's subcellular-localization assay (Figure 8) was performed **only** on GFP-tagged ARC, FAM40A, FAM40B, FMNL3, and ZRANB1. The verbatim passage reads: "*cDNAs for ARC, FAM40A, FAM40B, FMNL3 and ZRANB1 were cloned downstream of GFP and transfected into PC3 cells... showed cytoplasmic localization with some enrichment on the plasma membrane.*" C21orf2/CFAP410 was never among the tagged constructs and its localization was never directly observed. The only place "plasma membrane" is associated with C21orf2 in the paper is via the phenotype-class definition: "*cortical actin, increased F-actin on regions of the plasma membrane.*"

This is a classic curation-provenance failure: a **knockdown cortical-actin phenotype** (a functional/morphological readout) was mis-mapped to a **direct localization** of the knocked-down protein itself, and/or conflated with the GFP result for the five unrelated proteins. The

distinction the research question specifically flags — *cortical actin phenotype after CFAP410 knockdown* versus *localization of CFAP410 itself* — is exactly the axis along which this error occurred.

The 2024 Author Correction (P 39390519)

addresses only a duplicated Figure 6 image (a siARC panel that was identical to siZRANB1), states that the conclusions are unaffected, and makes no mention of C21orf2 or any localization result.

Defensible conclusion: CONTRADICTED. The GO:0005886 plasma-membrane IDA should be removed. Provenance class: mis-derived from a disease/morphology **phenotype**, not a **direct human localization assay**.

F002 — Localization consensus is the ciliary base; mitochondrial and outer-segment calls are weaker

QuickGO cellular-component provenance for O43822:

GO term	Compartment	Evidence	Source(s)	Species / basis
GO:0036064	Ciliary basal body	IDA	PMID: 26294103 (Khan 2015), PMID: 26167768 (Whewey 2015), HPA	Human/pig/mouse retina IHC + screen
GO:0032391	Photoreceptor connecting cilium	ISS	mouse ortholog Q8C6G1	Inferred from sequence/ortholog
GO:0001750	Photoreceptor outer segment	IDA	PMID: 27548899 (Suga 2016), IEA	Over-specified vs. "cilia"
GO:0005739	Mitochondrion	IDA	PMID: 9325172 (Krohn 1997), IEA	1997 polyclonal antisera
GO:0005829	Cytosol	IDA	HPA	Human Protein Atlas IF

Khan 2015 performed retinal immunohistochemistry across human, pig, and mouse and "*localised C21orf2 protein to the ciliary structures of the photoreceptor cell (the daughter basal body, the centriole adjacent to the basal body, and the connecting cilium)*" — explicitly **not** the

outer segment. Suga 2016's abstract states the protein "*localized to the photoreceptor cilia in the adult retina*," i.e. a ciliary compartment; the GO mapping to GO:0001750 "outer segment" is therefore **more specific than the paper supports**. De Decker 2025 (

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working in human iPSC-derived motor neurons, reported that "*C21ORF2 is located at the basal body of the primary cilium, and mutations associated with ALS alter this localization*" — independent, modern, direct human-cell evidence for basal-body localization.

The Krohn 1997 mitochondrial claim rests on 1997 polyclonal murine antisera (immunofluorescence with mitochondrial dye co-staining; a 25 kDa band on Western blot) with no knockout validation and no isoform-specificity control by modern standards.

Defensible conclusions: Ciliary basal body / connecting cilium / transition zone — **SUPPORTED** (multiple independent direct human assays across tissues and cell types). Photoreceptor outer segment — **CONTEXT-DEPENDENT / over-specified** (should be relabeled to the ciliary compartment the paper actually reports). Mitochondrion — **UNRESOLVED** (single low-confidence 1997 study; needs isoform-aware re-validation).

F003 — HPA data are "Supported" and place CFAP410 at Golgi/transition-zone/basal-body/cytosol; cytoplasm has a second, independent source

The Human Protein Atlas (ENSG00000160226, CFAP410) reports **Reliability (IF) = "Supported"** and **Reliability (IH) = "Supported."** Subcellular main location = Golgi apparatus, Primary cilium transition zone, Cytosol; additional location = Basal body. QuickGO maps the HPA IDA to GO:0005829 (cytosol) and GO:0036064 (ciliary basal body) via GO_REF:0000052. The basal-body and transition-zone HPA calls **corroborate** the independent ciliary-base localization from Khan 2015, Whewey 2015, and De Decker 2025.

Importantly, QuickGO shows the GO:0005737 "cytoplasm" IDA has **two** references:

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(Bai2011 — shown here to contain no CFAP410 localization assay) **and**

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(Fang 2015, "*The NEK1 interactor; C21ORF2, is required for efficient DNA damage repair*"). Because a second citation exists, removing the Bai2011 reference does **not** delete the cytoplasm term — but the Bai2011 citation on GO:0005737 is just as invalid as on GO:0005886 and should be stripped from both. Fang2015's full text was **not retrievable** through PMC efetch (only ~2.8

kB of front-matter returned; PMC4581587), so whether its cytoplasm assignment derives from direct IF, fractionation, or functional inference could not be verified in this audit. This is explicitly flagged as **inaccessible data**, not inferred.

F004 — All 14 Bai2011 supplementary files reviewed: none rescues the plasma-membrane IDA

Every supplementary file of Bai2011 (PMC3201212) was enumerated from the PMC XML and reviewed:

File	Content	Localization assay of C21orf2?
Additional file 1	Table S1 "Selection of PMMs" (gene names/domains/interaction partners, bioinformatics)	No
Additional file 2	Figure S1 (effects of PMM depletion on morphology — knockdown phenotype)	No
Additional file 3	Figure S2 "Examples of actin and shape phenotypes" (illustrative knockdown, ARC/FMNL1)	No
Additional file 4	Table S2 (phenotypes induced by individual siRNAs)	No
Additional files 5–12	Movies 1–8, time-lapse wound-healing (Control, ZRANB1, ARC, FMNL3, FNBP3, LIMD1, FAM40B, FAM40A)	No (none is C21orf2)
Additional file 13	Figure S3 (migration speeds)	No
Additional file 14	Table S3 (siRNA sequences)	No

No supplementary file reports any GFP or antibody localization of C21orf2/CFAP410 or any plasma-membrane assay of it. The **only** protein-localization experiment in the entire article remains main-text Figure 8 (GFP-ARC/FAM40A/FAM40B/FMNL3/ZRANB1). This closes the last plausible route by which the IDA could have been legitimately sourced. Fang2015 (PMC4581587) full text is likewise not available via Europe PMC (HTTP 404), confirming it is inaccessible for verification here.

F005 — UniProt's manual Subcellular Location comment omits plasma membrane; O43822 has 4 isoforms

The UniProt O43822 REST record (Cilia- and flagella-associated protein 410) lists a canonical sequence of **256 aa, 28,340 Da**, with **four annotated isoforms** (O43822-1 "Long" [displayed], O43822-2 "Short," O43822-3, O43822-4). The curated SUBCELLULAR LOCATION comment lists **only**:

- Mitochondrion (ECO:0000269, PubMed 9325172)
- Cytoplasm, cytoskeleton, cilium basal body (ECO:0000269, PubMed 26167768 + 26294103)
- Cell projection, cilium, photoreceptor outer segment (ECO:0000269, PubMed 27548899)
- Cytoplasm (ECO:0000269, PubMed 26290490)
- Note: "Colocalizes with NEK1 and SPATA7 at the basal body."

Plasma membrane is absent from the UniProt SL comment, and Bai2011

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is **not cited anywhere** in it — yet QuickGO shows both a GO:0005886 plasma-membrane IDA and a GO:0005737 cytoplasm IDA referencing

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(assignedBy UniProt/GOA). The plasma-membrane annotation is therefore an **isolated GOA annotation** not reflected in UniProt's manually curated localization section — the strongest internal-consistency argument for removal. The existence of four isoforms is also relevant to the older mitochondrial and antibody-based claims, none of which were isoform-resolved.

Mechanistic Model / Interpretation

The evidence converges on CFAP410 as a **ciliary-base protein** — concentrated at the basal body, daughter basal body, transition zone, and connecting cilium — consistent with its role as a NEK1/SPATA7 interactor at the ciliary base and its involvement in ciliopathies and ALS. The disputed and legacy annotations can be organized by evidence class:

CFAP410 (043822) localization – evidence hierarchy

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STRONGLY SUPPORTED -- Ciliary base -----
 Basal body / daughter basal body Khan2015 (human/pig/mouse retina IHC, IDA)
 Connecting cilium Wheway2015 (ciliopathy screen, IDA)
 Transition zone De Decker2025 (human iPSC motor neurons, IDA)
 HPA (IF "Supported": Golgi/TZ/basal body/cytosol)
 UniProt manual SL comment (+NEK1/SPATA7 note)

CONTEXT-DEPENDENT -- Photoreceptor "outer segment" (GO:0001750) -
 Suga2016 says "photoreceptor cilia" -> GO term over-specified;
 should map to ciliary compartment, not outer segment.

UNRESOLVED -- Mitochondrion (GO:0005739) -----
 Krohn1997: 1997 polyclonal antisera, 25 kDa band, mito-dye co-stain,
 no KO / no isoform control (4 isoforms exist). Needs re-validation.

CONTRADICTED -- Plasma membrane (GO:0005886) -----
 Bai2011 IDA: mis-derived from cortical-actin KNOCKDOWN phenotype
 (Table 2) and/or conflated with GFP result for 5 OTHER proteins
 (Fig 8: ARC/FAM40A/FAM40B/FMNL3/ZRANB1). CFAP410 never tagged.
 Absent from UniProt manual SL comment -> orphan GOA annotation.

The key mechanistic lesson for curation is that Bai2011 is a **cell-migration RNAi screen**, not a localization study of CFAP410. Its "plasma membrane" language belongs to (a) a phenotype-class label and (b) five unrelated GFP fusions. Annotation propagation transformed a phenotype into a localization — a misattribution that persists in GOA but was never accepted into UniProt's expert-curated SL section.

Evidence Base

Study	PMID	Role	Species / system	What it shows for CFAP410	Evidence class
Bai et al. 2011	21834987	Source of disputed IDA	PC3 cells, RNAi screen	C21orf2 = knockdown cortical-actin phenotype only (Table 2); never localized	Disease/morphology phenotype — not localization
Author Correction 2024	39390519	Correction to Bai2011	—	Fixes duplicated Fig 6 image only; C21orf2 untouched	Correction, not relevant to claim
Khan et al. 2015	26294103	Primary localization	Human/pig/mouse retina IHC	Daughter basal body, adjacent centriole, connecting cilium — NOT outer segment	Direct assay (multi-species)
Wheway et al. 2015	26167768	Primary localization	Ciliopathy screen	Ciliary basal body	Direct assay
Suga et al. 2016	27548899	Primary localization	Retina	"photoreceptor cilia" (abstract) → GO "outer segment" is over-specified	Direct assay (compartment over-mapped)
De Decker et al. 2025	39703094	Primary localization	Human iPSC motor neurons	Basal body of primary cilium; ALS mutations alter localization	Direct human-cell assay
Fang et al. 2015	26290490	Second cytoplasm source	—	NEK1 interactor required for DNA-damage repair; cytoplasm basis unverified	Inaccessible (full text 404/ front-matter only)

Study	PMID	Role	Species / system	What it shows for CFAP410	Evidence class
Krohn et al. 1997	9325172	Mitochondrion source	Polyclonal antisera	25 kDa band, mito co-stain; no KO/isoform control	Legacy direct assay, low confidence
Endogenous complex 2023	37188479	Endogenous complex	—	NEK1–C21ORF2 association; supports no MF/ localization claim alone	Association only
Neuronal study 2025	39227882	Neuronal context	—	Neuronal/ALS context; no independent localization claim adopted here	Secondary / context
DS brain study	15068244	Secondary discussion	DS/AD frontal cortex	C21orf2 reduced in Down syndrome; speculates mitochondrial dysfunction	Secondary/ disease context

The Human Protein Atlas (ENSG00000160226; IF and IH reliability both "Supported") provides an independent, systematic corroboration of the ciliary-base and cytosol/Golgi/transition-zone calls, and is the source for the HPA-derived GO:0005829 and GO:0036064 IDAs.

Claim-by-Claim Curation Table

GO term	Compartment	Primary evidence	Evidence class	Counterevidence	Conclusion
GO:0005886	Plasma membrane	Bai2011 IDA (P 21834987)	Phenotype mis-mapped; NOT a localization assay	Full text + 14 supplements show no CFAP410 localization; absent from UniProt SL	CONTRADICTED — remove
GO:0005737	Cytoplasm	Bai2011 (invalid) + Fang2015 (P 26290490)	IDA; second source unverifiable	Bai2011 citation invalid; Fang2015 inaccessible	CONTEXT-DEPENDENT — strip Bai2011, retain under Fang2015 pending verification
GO:0036064	Ciliary basal body	Khan2015, Wheway2015, De Decker2025, HPA	Direct human/ multi-species IDA	None	SUPPORTED
GO:0032391	Connecting cilium	Mouse ortholog Q8C6G1 (ISS) + Khan2015	Ortholog inference + direct	None	SUPPORTED
GO:0001750	Photoreceptor outer segment	Suga2016 (P 27548899)	Direct assay, compartment over-mapped	Paper says "photoreceptor cilia"; Khan2015 excludes outer segment	CONTEXT-DEPENDENT — re-map to ciliary compartment
GO:0005739	Mitochondrion	Krohn1997 (P 9325172)	1997 polyclonal antisera	No KO/isoform control; 4 isoforms exist	UNRESOLVED — re-validate
Kinase activator /	—		Association only	No functional assay	NOT SUPPORTED — do not infer

GO term	Compartment	Primary evidence	Evidence class	Counterevidence	Conclusion
scaffold (MF)		NEK1 association (P 37188479)			from association

Limitations and Knowledge Gaps

1. Fang2015

([P 26290490](#))

full text is inaccessible. Both PMC efetch (only ~2.8 kB front-matter, PMC4581587) and Europe PMC (HTTP 404) failed to return the article. The exact basis of the GO:0005737 "cytoplasm" IDA attributed to this paper (direct IF vs. fractionation vs. functional inference) therefore **could not be verified**. This is flagged as inaccessible data, not inferred absence.

2. **Isoform resolution is missing throughout the legacy record.** O43822 has four isoforms, but none of the mitochondrial (Krohn1997), antibody-based, or older annotations distinguished isoform-specific targeting. Antibody cross-reactivity and isoform-specific epitopes remain unaddressed.

3. **Mitochondrial claim relies on 1997 polyclonal antisera** without knockout controls; it cannot be confidently accepted or rejected with current data.

4. **Photoreceptor outer-segment GO mapping is over-specified.** Suga2016's own wording ("photoreceptor cilia") does not license the outer-segment (GO:0001750) term specifically; the mapping conflates ciliary compartments.

5. **Provenance tracing is partial.** QuickGO confirms the Bai2011 IDA is "assignedBy UniProt," but the precise internal curation step that converted a knockdown phenotype into a localization annotation could not be reconstructed from public records.

Proposed Follow-up Experiments / Actions

1. **Curation action (immediate):** Remove the GO:0005886 plasma-membrane IDA and strip the

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citation from the GO:0005737 cytoplasm IDA for O43822. File a GOA annotation-quality report referencing this provenance audit.

2. **Verify Fang2015:** Obtain the publisher PDF/supplements of

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(institutional or interlibrary access) to confirm whether its cytoplasm assignment is a direct IF observation or a functional inference; adjust the GO:0005737 evidence code accordingly.

3. **Isoform-resolved localization:** Express each of the four O43822 isoforms as endogenous-level tagged constructs (CRISPR knock-in) in ciliated human cells (e.g., RPE1) and image against basal-body/transition-zone markers; specifically test whether any isoform shows mitochondrial targeting to adjudicate Krohn1997.
4. **Antibody validation with knockout controls:** Re-run the mitochondrial and outer-segment immunostaining using CFAP410-knockout cells/tissue as a negative control and validated isoform-specific antibodies to resolve cross-reactivity.
5. **Compartment re-mapping:** Recommend to GO/UniProt that the Suga2016 annotation be re-mapped from GO:0001750 (outer segment) to the ciliary compartment the paper reports ("photoreceptor cilia").

6. **Endogenous complex context:** Treat

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(NEK1–C21ORF2 complex) as evidence of physical association only; do not derive kinase-activator, scaffold, or localization terms from it without direct functional assays.

Prepared as a targeted primary-evidence audit for GO curation. Direct human assays, ortholog/ISS inferences, predictions, secondary discussion, and disease phenotypes have been kept distinct throughout; inaccessible sources (Fang2015 full text) are explicitly flagged rather than inferred from abstract absence.

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